

Nmap – Further Nmap

TryHackMe Practical Cybersecurity Lab Documentation

Focus: Network scanning, port states, scan types, UDP scanning, NSE, host discovery and firewall considerations.

Lab Platform: TryHackMe

Room: Further Nmap

Completion: 15 tasks • 328 points



Nmap completed!

subhashini22 completed another room on their cyber security journey.

Completed tasks	Points earned	Streak
 15	 328	 1

Figure 1 – TryHackMe Further Nmap completion (15 tasks, 328 points)

1. Overview

I completed the TryHackMe **Further Nmap** room to build practical knowledge of Nmap scanning and enumeration. The lab covered different TCP and UDP scan techniques, port states, host discovery, timing and verbosity options, output formats, and the Nmap Scripting Engine (NSE).

2. Key Concepts Learned

Topic	What I practiced / understood
TCP scanning	TCP Connect (-sT) and SYN (-sS) scans
UDP scanning	UDP scan (-sU) and UDP port interpretation
Port states	Open, closed and filtered behavior
Special TCP scans	NULL (-sN), FIN (-sF) and Xmas (-sX)
Host discovery	Ping sweep with -sn and bypassing discovery with -Pn
Scanning options	Verbosity (-v/-vv), timing templates (-T0 to -T5), -p-
NSE	Script execution, vulnerability scripts and script discovery
Enumeration	FTP anonymous access and SMB OS discovery
Firewall considerations	How filtering and packet behavior can affect results

3. Practical Scan – TCP

Authorized TryHackMe target used in the lab:

```
nmap 10.49.190.20
```

The scan identified the following open TCP services and 996 closed TCP ports:

Port	State	Service
21/tcp	Open	FTP
22/tcp	Open	SSH
53/tcp	Open	DNS
80/tcp	Open	HTTP

```

root@ip-10-49-134-203:~# nmap 10.49.190.20
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-16 05:38 UTC
Nmap scan report for ip-10-49-190-20.ap-south-1.compute.internal (10.49.190.20)
Host is up (0.00025s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
MAC Address: 0A:FF:C9:1E:26:B1 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds

```

Figure 2 – Nmap TCP scan result from the authorized TryHackMe lab

4. Practical Scan – UDP

I also tested UDP port 161, commonly associated with SNMP:

```
nmap -sU -p 161 10.49.190.20
```

The result showed **161/udp closed snmp**. This demonstrates that UDP scanning uses different response behavior from TCP scanning.

```

root@ip-10-49-134-203:~# nmap -sU -p 161 10.49.190.20
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-16 05:41 UTC
Nmap scan report for ip-10-49-190-20.ap-south-1.compute.internal (10.49.190.20)
Host is up (0.00011s latency).

PORT      STATE SERVICE
161/udp    closed snmp
MAC Address: 0A:FF:C9:1E:26:B1 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds

```

Figure 3 – Nmap UDP scan of port 161

5. Important Nmap Commands

TCP Connect scan

```
nmap -sT <TARGET-IP>
```

SYN scan

```
nmap -sS <TARGET-IP>
```

UDP scan

```
nmap -sU <TARGET-IP>
```

NULL / FIN / Xmas scans

```
nmap -sN <TARGET-IP>
```

```
nmap -sF <TARGET-IP>
```

```
nmap -sX <TARGET-IP>
```

Host discovery

```
nmap -sn 172.16.0.0/16
```

Skip host discovery

```
nmap -Pn <TARGET-IP>
```

Scan all TCP ports

```
nmap -p- <TARGET-IP>
```

Run vulnerability NSE scripts

```
nmap --script vuln <TARGET-IP>
```

FTP anonymous enumeration

```
nmap -p 21 --script ftp-anon <TARGET-IP>
```

6. Nmap Scripting Engine (NSE)

NSE extends Nmap beyond basic port scanning by allowing scripts to perform additional enumeration and security checks. During the room, I learned how to search for installed scripts, use vulnerability scripts, identify SMB-related scripts, and test FTP anonymous access.

Example:

```
nmap --script http-title <TARGET-IP>
```

7. Nmap Timing and Output

Nmap provides timing templates from **-T0** (paranoid) through **-T5** (insane). Verbosity can be increased with **-v** or **-vv**. Scan results can also be saved using **-oN**, **-oX**, **-oG**, or **-oA**.

8. Key Takeaways

- Nmap is more than a basic port scanner; its output needs to be interpreted in context.
- Different scan types produce different network behavior and results.
- UDP scanning requires different interpretation because UDP does not use a TCP three-way handshake.
- Host discovery can be affected by firewalls and filtering; **-Pn** can be useful when discovery probes fail.
- NSE scripts provide additional enumeration and security-checking capabilities.
- Scanning should be performed only against systems where authorization has been provided.

9. Completion

TryHackMe room: Further Nmap

Completed tasks: 15

Points earned: 328

Tools: Nmap, Nmap Scripting Engine (NSE), Linux terminal, TryHackMe

10. Disclaimer

All scans documented in this report were performed against an authorized TryHackMe lab environment for educational purposes. The commands and techniques should only be used on systems for which you have explicit authorization.